

FOC Tuning Guide

Power Application Controller®



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1 INITIAL SETUP

1.1 Select Compile Options

Manual tuning needs to use the Align&Go startup mechanism. Therefore the procedure outlined in this guide requires that the firmware be configured to use the Estimator module with Align&Go startup.

The Estimator module with Align&Go startup is enabled in “config_app.h” by selecting the following MPOS mode definition:

- 1) **#define** MPOS_SELECT MPOS_LBG1_START_ALIGNGO
This MPOS option starts the motor using Align&Go and switches to sensorless mode when the speed is above a user-configurable threshold.

1.2 Initial GUI Parameter Values

Please follow the steps below to configure the required GUI parameters before starting the tuning process:

- 1) Load the XML file into the GUI called “start_foc_tuning_gui_settings.xml”. This file was provided along with the GUI and contains the default values for getting started with the tuning process.
- 2) Configure the values in the Motor Config section:
 - a. $L_d / L_q / R_{\alpha}$ ($\mu\text{H} / \mu\text{H} / \text{Ohms}$)
 - When measuring winding inductance using an LCR meter, measure the inductance at various rotor positions as this value might change depending on the relative position between the magnets and the phase windings. The largest inductance value is typically entered as L_q and the smallest entered as L_d .
 - If these values are not available please refer to the auto-tuning document to obtain them.
 - b. Pole Pairs
 - If the number of pole pairs is not known, it can be determined later in the tuning process. Refer to Section 3.2 for more information on determining the number of pole pairs.
 - This value is only necessary for auto-tuning the speed controller and for calculating speed in RPM in the GUI. If doing manual tuning these values and RPM display is not critical this value can be left as 0.
 - c. K_e ($\text{V} \cdot \text{sec} / \text{rad}$)
 - This value is necessary for auto-tuning the speed controller and when either Ke-based v-SIM or i-SIM are enabled. If neither of these features are used this value can be left as 0.
 - d. J (Nm / a)
 - This value is only necessary for auto-tuning the speed controller. If doing manual tuning this value can be left as 0.
 - e. Nominal Speed (Hz)
 - This is the expected maximum speed that the motor will run at under normal conditions. This information is used to calculate the suggested bandwidth when using the sensorless rotor position estimator.
- 3) Configure the values in the Control Config section:
 - a. PWM Frequency (Hz)
 - Typically a value of 20kHz is adequate for most applications.
 - b. Control Divider
 - Sets the control frequency as a function of PWM frequency :
$$f_{\text{control}} = f_{\text{pwm}} / \text{Divider}$$
 - Typically a value of 1 when setting the PWM frequency to 20kHz is adequate for most applications.
- 4) Configure the values in the App Config section:
 - a. Max Speed (Hz)
 - This value is the maximum speed allowed before the motor speed fault triggers. It also used to determine an adequate speed scale for the application.
 - b. Max Power (W)
 - This value is only required when choosing power as the outerloop control method. If power control is not selected, this value can be 0.
 - c. Max Current (A)
 - Enter the maximum phase current allowed by the current controllers during high speed or maximum acceleration.
 - d. Negative Torque (A)
 - Enter the maximum phase current allowed by the current controllers while braking or decelerating.
 - e. Over-Current Limit (A)

- Enter the maximum phase current allowed by the integrated hardware current limit that shuts down the motor for protection.
- When the Cycle-by-Cycle (CBC) current limiting mode is enabled, it uses the current limit value as the CBC limit and the shutdown limit for catastrophic failures becomes 2X this current limit. When CBC current limiting mode is disabled, the current limit value becomes the shutdown limit.
- f. Coast Timeout (s)
 - Enter the time after which the firmware will assume that the motor is stopped during coasting. This is used to ensure that the firmware always can exit the coasting state even if it is unable to detect that the motor is stopped due to noise.
 - Recommended value: 2x the time it takes for motor to free spin to a stop
- 5) Configure and verify the values in the Hardware Config section:
 - a. DiffAmp Gain
 - Usually a value of 4x is preferred, assuming that an adequate value of sense resistor is chosen.
 - Typically the sense resistor is chosen such that when the motor current reaches the Over-Current Limit, the value across the sense resistor does not exceed +/-0.3V since this is the limitation of the differential amplifier inputs:

$$OCLimit * Rsense \leq 0.3V$$
 - b. OC Rsense / Iuvw Rsense (mOhms)
 - Check that over-current and UVW-phase current sense resistor values match the values on the PCB.
 - c. Vphase Ratio
 - Verify that the bus voltage sensing resistor ratio matches the values on the PCB.
- 6) Configure the Current Control section:
 - a. Iq/I_d K_p = 0
 - b. Iq/I_d K_i = 0
 - c. I_d & I_q ramp rate = 10% of Max Current or the lowest value allowed by GUI
- 7) Configure the Speed Control section:
 - a. Speed K_p = 0
 - b. Speed K_i = 0
 - c. Speed ramp rate = 10% of Max Speed or the lowest value allowed by GUI
- 8) Configure the Operation section:
 - a. CL Mode
 - Select Torque Mode.
 - b. Speed setpoint (Hz)
 - Set to same value as Align&Go “OL speed setpoint”. If this value has not been determined yet, this can initially be set to 0. When the Align&Go section is tuned, this value should also be set.
 - c. I_d ref CL (A)
 - Should always be set to 0 for tuning.
 - d. I_q ref CL (A)
 - Set to 10% of Max Current.

2 CONTROLLER TUNING

2.1 Background and Considerations

Some basic concepts of PI controllers are necessary for proper tuning:

Kp Gain

- Refers to the gain of the proportional term of the controller output
- Responsible for the immediate response to changes in the controller inputs
- P-term becomes less dominant as the feedback approaches the target

Ki Gain

- Refers to the gain of the integral term of the controller output
- I-term takes over for finally converging the feedback to the target after the error becomes smaller and the P-term becomes less dominant

The table below summarizes the typical characteristics of the step response of improperly tuned PI controllers.

PI Parameter	Low	High
Kp	Possibly unstable or overshoot	• “spike” overshoot • possible audible high frequency noise due to an output that is too responsive to noise on the feedback signal.
Ki	May not converge to the commanded current	Possibly unstable or overshoot

Table 1: Typical Characteristics of Improperly Tuned PI Controllers

2.2 Auto Tuning (Recommend)

The FOC project includes a tool for obtaining the motor parameters and also for automatically tuning the current and speed controllers. **This is a recommended approach to tune a new motor.** Please refer to the Auto-Tuning Guide (..\resources\guides\features) for instructions on how to use it.

Sometimes it may be necessary to tune the controllers manually:

- The auto-tuning tool may sometimes have some trouble correctly tuning certain motors.
- The load attached to the motor may not allow the user to properly tune the speed controller.
- The user may want to go through the manual process even after running auto-tuning in order to verify that the tuning is adequate or if fine tuning is necessary

If manual tuning is desired please follow the procedure described below in Section 2.3

2.3 Manual Tuning

This section describes the procedure to manually tune the current and speed controllers.

2.3.1 Current Controllers

Current controller tuning involves 2 basic steps:

- 1) Making sure that the motor current converges to the defined target
- 2) Making sure that the step response is stable

2.3.1.1 Tuning Current Convergence

Please follow the steps below to achieve convergence with the current controllers:

- 1) Check the following settings in the GUI
 - a) Verify that the Close Loop (Auto) box is unchecked to stay in open-loop mode.
 - b) Set Align&Go OL speed setpoint = 0 so that the motor driver just applies a constant current to the phases.
 - c) Verify that the Id/IqRef ramp rate to is set to a low value, perhaps 1A/s or whichever rate is the minimum the GUI will allow.
- 2) Set the PWM DAC to look at the Iq controller values. The signals to be displayed are listed below:
 - a) DAC1 = Iq Reference
 - b) DAC2 = Iq Feedback
 - c) DAC3 = Iq PI output

Note: The PWM DAC scale factors may need to be adjusted starting with a value of 1.
(Tools -> PWM DAC Scale Factors)

Note: If the DAC's are not available for the tuning process, the phase current waveform by itself can be used instead to represent the behavior of the Iq controller.

- 3) Enable the motor and observe the Iq waveform on the PWM DAC. Iq should converge to Iq_ref. If it does not converge:
 - a) Increase Ki gain if there is a constant "DC" difference/separation
 - b) Increase Kp gain if Iq is not very flat. It is OK to have some fluctuation.
 - c) If the current regulation is not stable, try reducing both Kp and Ki gains.



- 4) Repeat the same steps for the Id Controller.

Note: If the PWM DAC's are not available, the same gains applied to Iq can also be applied to Id.

2.3.1.2 Tuning Current Step Response

Please follow the steps below to ensure the stability of the current controller's step response:

- 1) Check the following settings in the GUI
 - a) Verify that the Close Loop (Auto) box is unchecked to stay in open-loop mode.
 - b) Set Align&Go OL speed setpoint = 0 so that the motor driver just applies a constant current to the phases.
 - c) Change the Id/IqRef ramp rate to 1000A/s to simulate step function when the reference is changed.
- 2) Set the PWM DAC to look at the Iq controller values. The signals to be displayed are listed below:
 - a) DAC1 = Iq Reference
 - b) DAC2 = Iq Feedback
 - c) DAC3 = Iq PI output

Note: If the DAC's are not available for the tuning process, the phase current waveform by itself can be used instead.

- 3) Change the value of Align&Go "I total ref OL" and check that the response is quick but stable.
- 4) If the response is not stable:
 - a) Increase Kp and/or
 - b) Decrease Ki
- 5) After changing the gains in Step#4, it is necessary to verify again that the controller is able to converge to the target value.
- 6) Do the same for the Id controller. The gain values obtained for the Iq controller can be used as a starting point. The Id current target can be adjusted by leaving "I total ref OL" unchanged and changing "Id ref OL" between 0 and a value equal to the value in "I total ref OL"

Note: If the PWM DAC's are not available, the same gains applied to Iq can also be applied to Id.

- 7) Restore the Id/IqRef ramp rate back to the desired ramp rate.

2.3.1.3 Tuning Current Response in Closed-Loop

Although the step response of the current controllers has already been evaluated in the previous section, this was done while the motor was not turning and the system was still in open loop (using a forced angle). The step response should be looked at in more detail in closed loop (using the actual motor angle information) to ensure that the response is well-behaved.

- 1) The MPOS method used must to be tuned since this procedure requires that the motor is running in closed loop. Please refer to the relevant MPOS Guide for instructions on how to do this before proceeding.
- 2) Enable the motor in Torque Mode and ensure that it is running in closed-loop mode.
- 3) Repeat the steps listed in Section 2.3.1.2 .
- 4) Adjust the controller gains as necessary to obtain the desired response and rise time. Please use the following scope shots in Figure 1, Figure 2, and Figure 3 as a guide. Some general guidelines:
 - a. If the response is not stable, increase K_p and/or decrease K_i
 - b. If the response is too slow, increase K_p and/or K_i

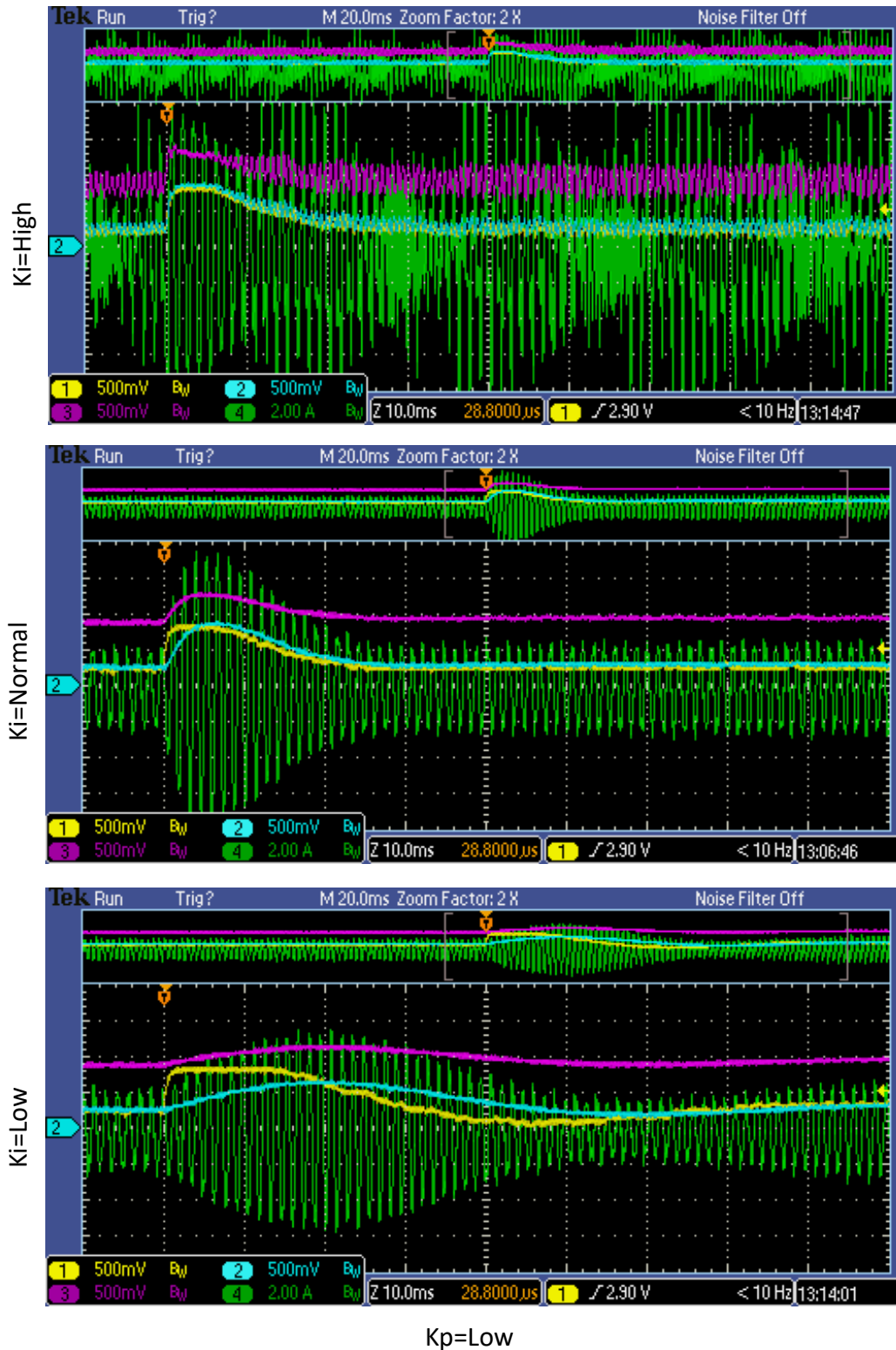
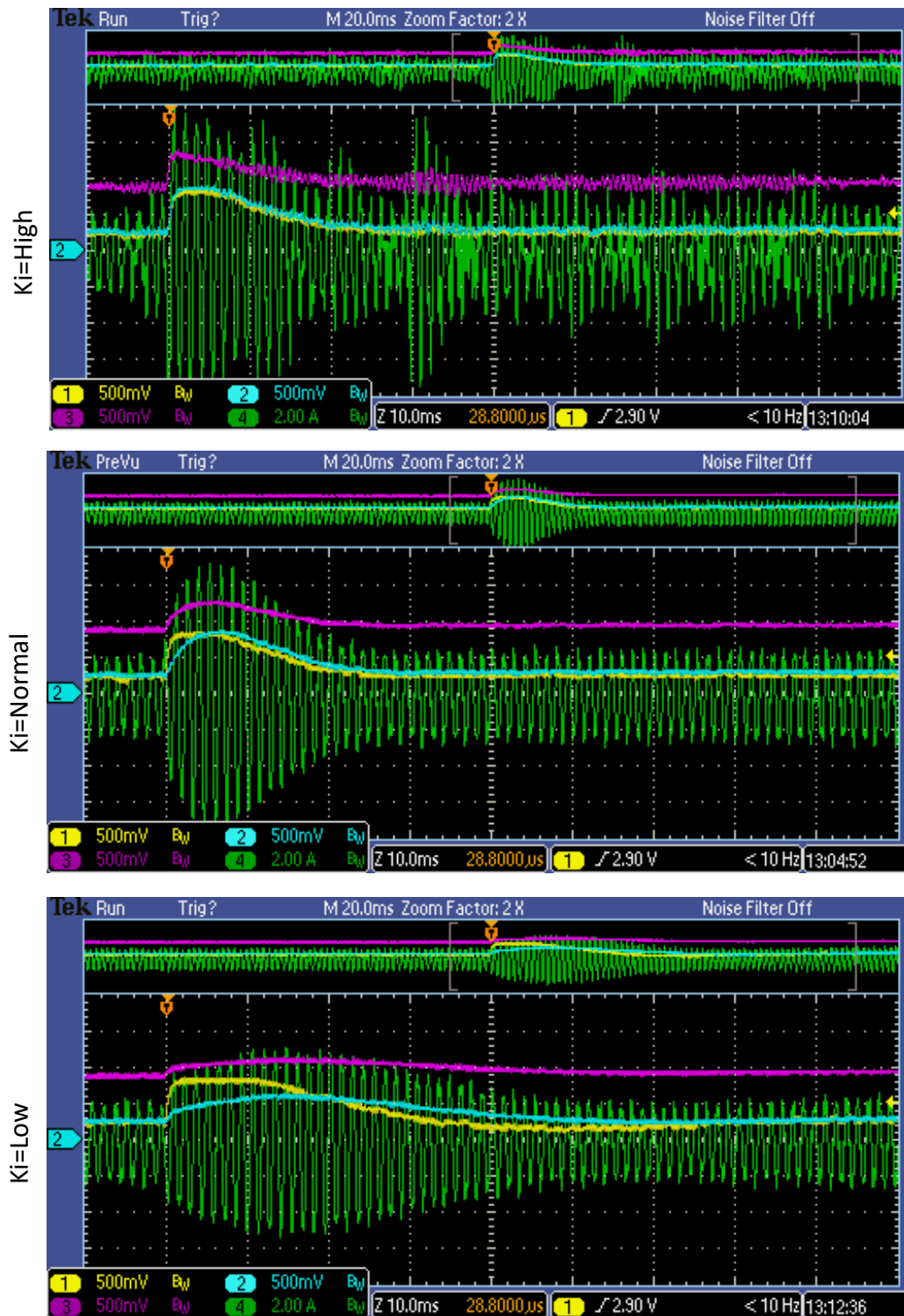


Figure 1: Example of Current Controller Step Responses - Kp Low



Kp=Normal

CH1=Speed Ref / CH2=Est Speed / CH3=Iq Ref / CH4=Iphase

Figure 2: Example of Current Controller Step Responses - Kp Normal

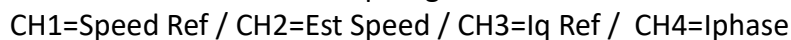


Figure 3: Example of Current Controller Step Responses - Kp High

2.3.2 Speed Controller

Similarly to the current controller, the speed controller tuning involves 2 basic steps:

- 1) Making sure that the motor speed converges to the defined target
- 2) Making sure that the step response, or the fastest desired motor response, is stable

2.3.2.1 Tuning Speed Convergence

Please follow the steps below to achieve convergence with the speed controller:

- 1) Check the following settings in the GUI
 - a) Verify that the Close Loop (Auto) box is checked to enter in closed-loop mode.
 - b) Verify that the closed-loop speed ramp rate is set to a low value, perhaps 10Hz/s or whichever rate is the minimum the GUI will allow.
 - c) Verify that the MPOS speed filter is set to a cutoff frequency that is adequate for the desired acceleration rates. For acceleration in the 10-50ms range, a cutoff frequency as high as 100Hz may be necessary. For most other slow acceleration applications 20Hz is adequate.
- 2) Set the PWM DAC to look at the Speed controller values. The signals to be displayed are listed below:
 - a) DAC1 = Speed Reference
 - b) DAC2 = Speed Feedback
 - c) DAC3 = Speed PI output

Note: The PWM DAC scale factors may need to be adjusted starting with a value of 1.
(Tools -> PWM DAC Scale Factors)

Note: If the DAC's are not available for the tuning process, the Speed and Target Speed displays can be used to judge the convergence of the speed controller.

- 3) Enable the motor and observe the speed waveform on the PWM DAC. Speed Feedback should converge to Speed Reference. If it does not converge:
 - a) Increase Ki gain if there is a constant "DC" difference/separation
 - b) Increase Kp gain if Iq is not very flat. It is OK to have some fluctuation.
 - c) If the speed regulation is not stable, try reducing both Kp and Ki gains.
- 4) Repeat the same steps at different speeds. By the end of the speed convergence tuning, a sweep through the entire speed range should complete without any issues as shown in the plot in Figure 4.



CH1=Speed Setpoint / CH2=Estimated Speed / CH4=Iphase

Figure 4: Speed Sweep after Convergence Tuning

2.3.2.2 Tuning Speed Step Response

When fast acceleration is required, a quick increase in motor current is required to provide the necessary torque. Therefore when the speed command changes suddenly, the speed controller will also suddenly change the I_q current reference. As a result the current controllers must be able to respond very quickly to this change.

Please follow the steps below to ensure the stability of the speed controller's step response:

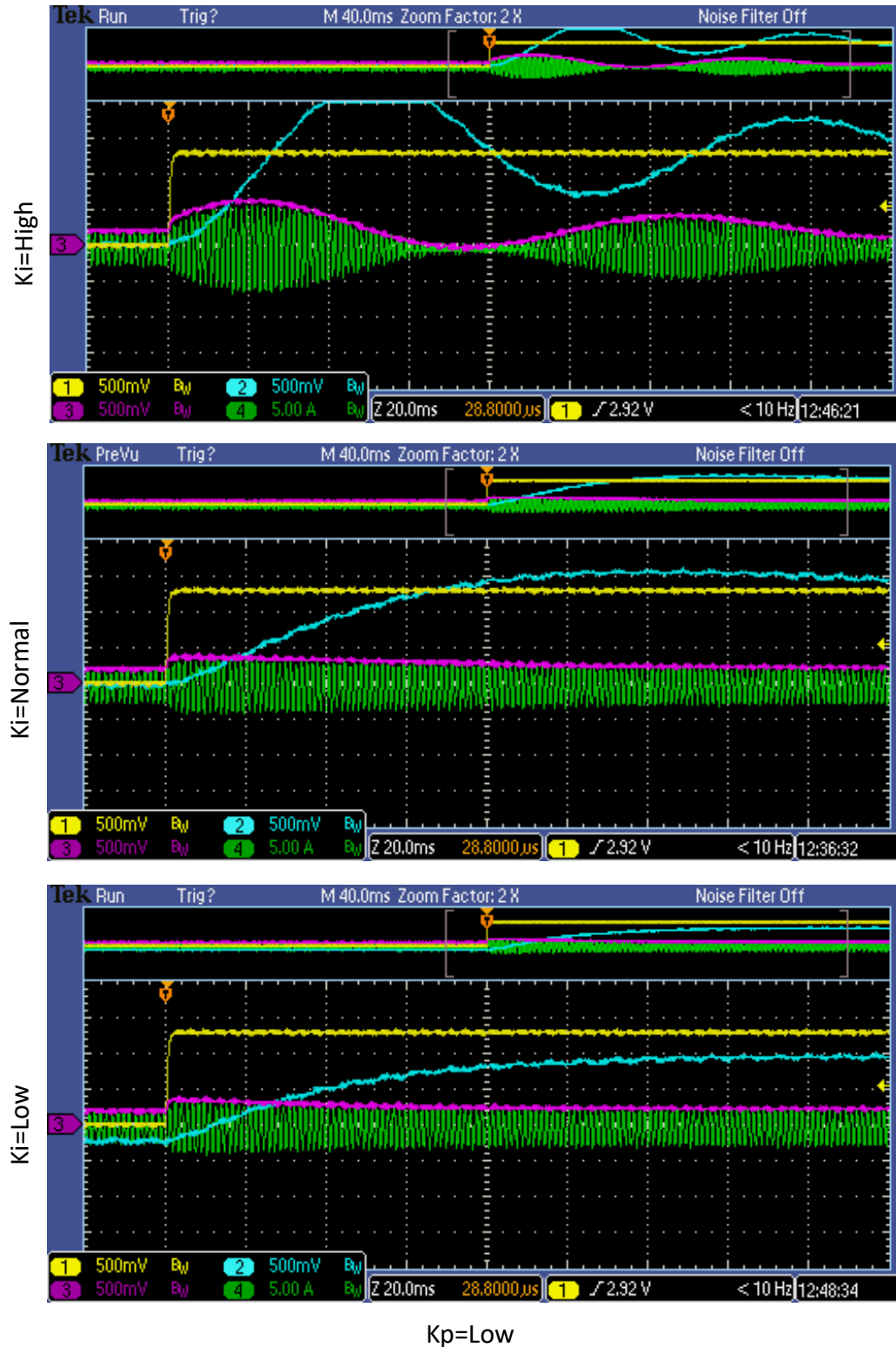
- 1) The Align&Go startup and the Estimator need to be tuned since this procedure requires that the motor is running in closed loop. Please refer to MPOS Estimator Guide for instructions on how to do this before proceeding.
- 2) Verify that the current controllers are properly tuned according to the previous sections dedicated to the current controller tuning.
- 3) Check the following settings in the GUI
 - a) Verify that the motor is running in closed-loop and set to Speed Mode.
 - b) Set "CL speed ramp" rate to a high number such as 400000Hz/s to simulate step function when the reference is changed.
- 4) Set the PWM DAC to look at the Speed controller values. The signals to be displayed are listed below:
 - a) DAC1 = Speed Reference
 - b) DAC2 = Speed Feedback
 - c) DAC3 = Speed PI output

Note: If the DAC's are not available for the tuning process, the envelope of the phase current waveform by itself can be used instead as an indicator of speed response stability.

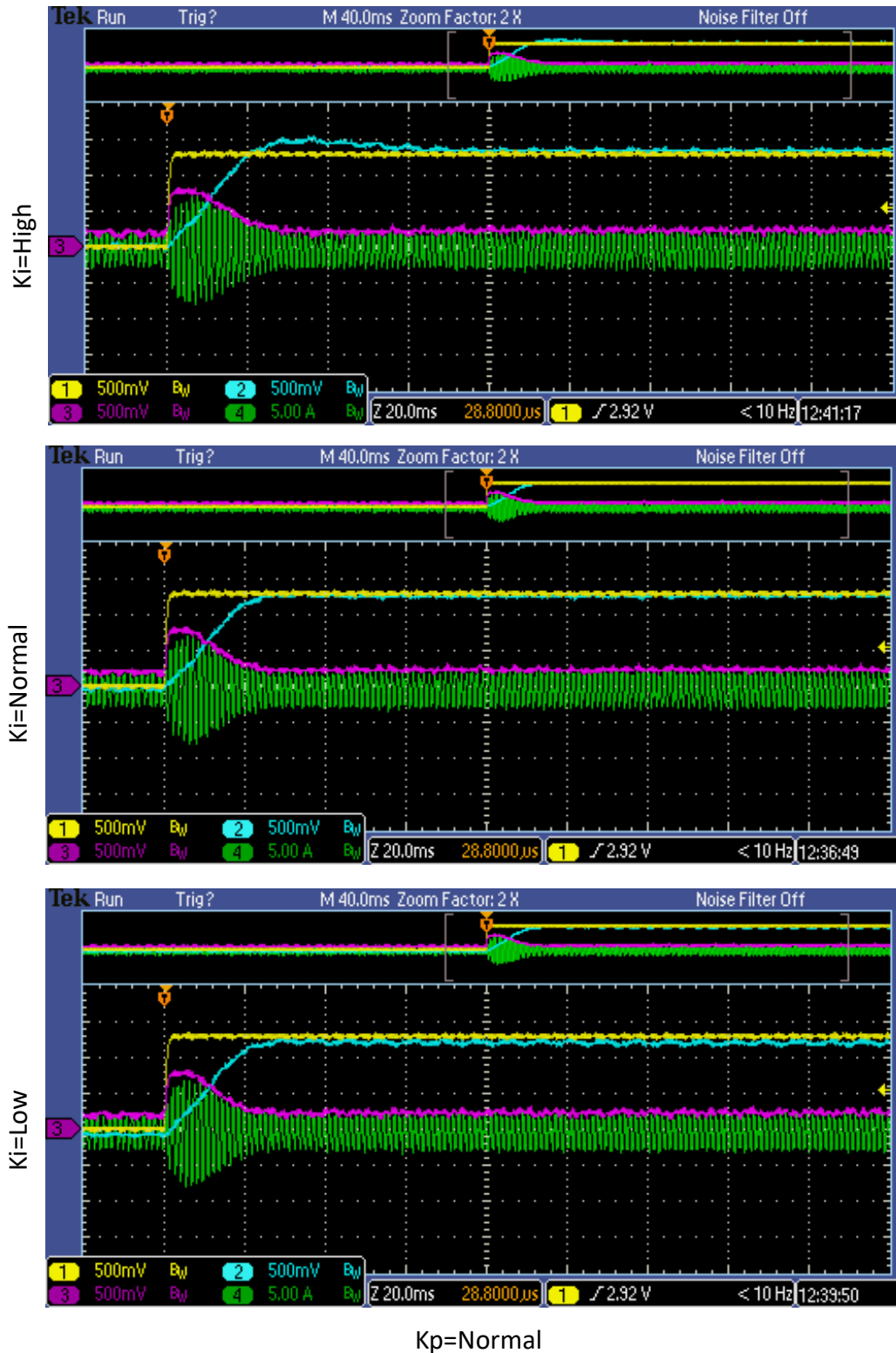
- 5) With the motor running in closed-loop mode and in Torque Mode, change the value of Operation "Speed setpoint" and check that the response is quick but stable.
- 6) Adjust the controller gains as necessary to obtain the desired response and rise time. Please use the following scope shots in Figure 5, Figure 6, and Figure 7 as a guide. Some general guidelines:
 - a) If the response is not stable, increase K_p and/or decrease K_i
 - b) If the response is too slow, increase K_p and/or K_i

Note: The saturation limit of the speed controller can affect the maximum acceleration rate that the motor can achieve, especially for large speed steps or when running at high speed. Please make sure that this value is appropriate for the acceleration rates necessary in the application

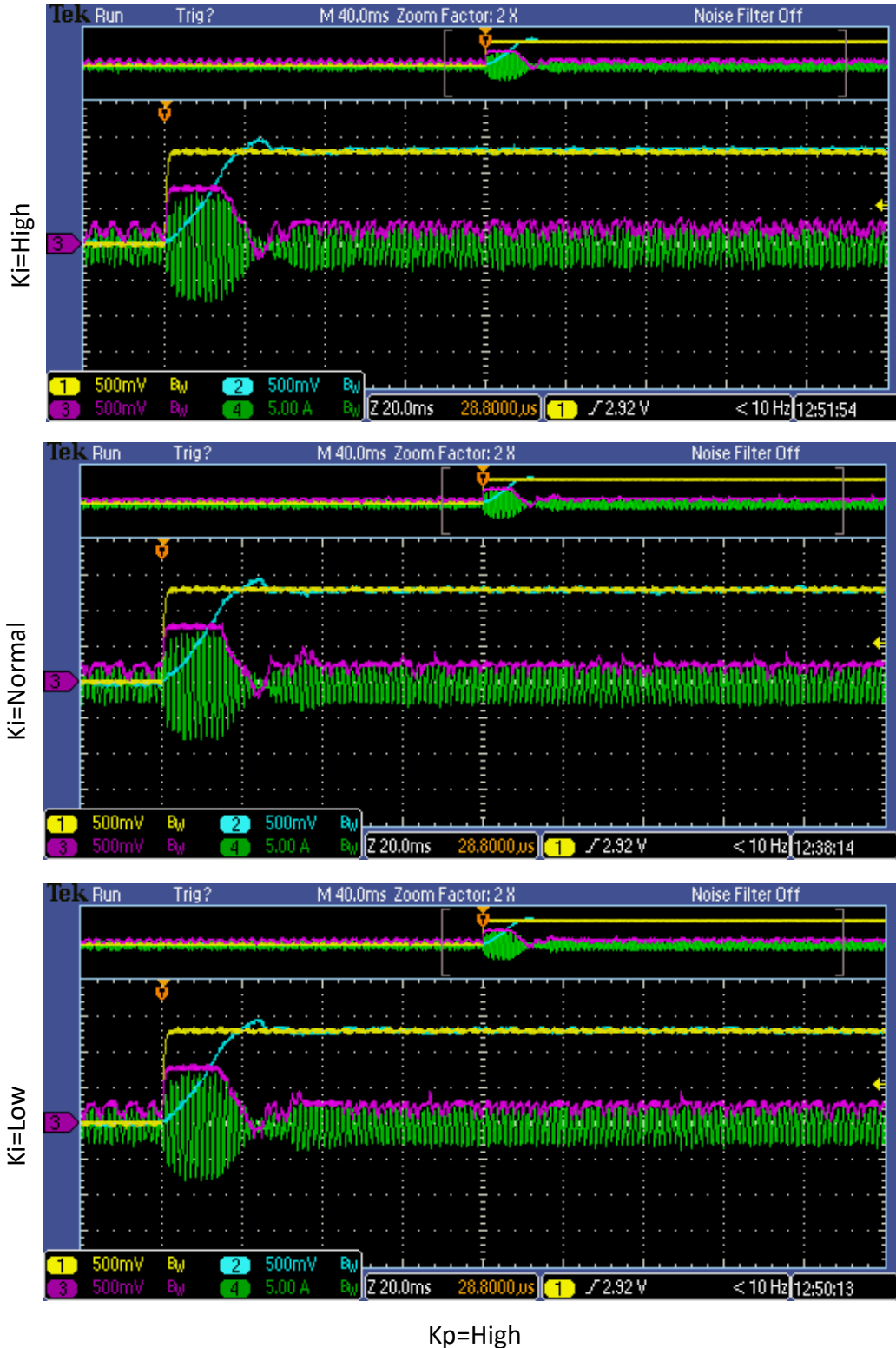
- 7) After changing the gains in Step#6, it is necessary to verify again that the controller is able to converge to the target value.
- 8) Restore the "CL speed ramp" rate back to the desired ramp rate.



CH1=Speed Ref / CH2=Est Speed / CH3=Iq Ref / CH4=Iphase
Figure 5: Example of Speed Controller Step Responses - Kp Low



CH1=Speed Ref / CH2=Est Speed / CH3=Iq Ref / CH4=Iphase
Figure 6: Example of Speed Controller Step Responses - Kp Normal



Kp=High
CH1=Speed Ref / CH2=Est Speed / CH3=Iq Ref / CH4=Iphase
Figure 7: Example of Speed Controller Step Responses - Kp High

3 MISCELLANEOUS INFO

3.1 General Troubleshooting

If the motor stalls when accelerating quickly, it is possible that the following parameters may need to be adjusted:

- 1) Estimator bandwidth gain may need to be decreased in order to increase the ratio of control frequency to estimator bandwidth.
- 2) Further Id PI controller tuning may be necessary to ensure that the controller is responding fast enough to the fast rotor angle change.
- 3) Control frequency may need to be increased especially to be able to track fast changes in speed while operating at high speed.

3.2 Determining Pole Pairs

In order to determine the number of pole pairs of a motor please follow the following steps:

- 1) Verify that the Align&Go startup is configured according to MPOS Estimator Guide.
- 2) Check the following settings in the GUI
 - a. Verify that the Close Loop (Auto) box is unchecked to stay in open-loop mode.
 - b. Set Align&Go OL speed setpoint = 1 so that the motor is driven at 1Hz.
- 3) Enable the motor and measure the time it takes for the motor to complete a mechanical revolution.
- 4) Calculate the number of pole pairs as follows:

$$\text{Pole Pairs} = \text{Mechanical Period (s)} / \text{Electrical Period (s)}$$

3.3 Troubleshooting on Failures during compiling and powerup

During compile time, flash overflow may happen if enabling too many features at the same time. For example, “ENABLE_AUTOTUNE_MID” on PAC52xx with Eclipse may incur flash overflow. In this case, please disable any unnecessary features to the motor parameter identification, such as:

```
// #define ENABLE_IPD
// #define ENABLE_SIM
// #define ENABLE_PLL_ANGLE_CALCULATION
// #define ENABLE_ANGLEDIFF_AUTORANGE
```

When “AFE not initialized properly” is displayed on the very first powerup, please double check if the correct board file is selected from “config_board.h”.



CONTACT INFORMATION

For the latest specifications, additional product information, worldwide sales and distribution locations:

Web: www.qorvo.com

Tel: 1-844-890-8163

Email: customer.support@qorvo.com

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